HOSTEL ROOM ISSUE TRACKER

A PROJECT REPORT

Submitted by

M. SIVA SWATHI

ROLL NO. 240701516

in partial fulfilment of the End Semester Examination for the course

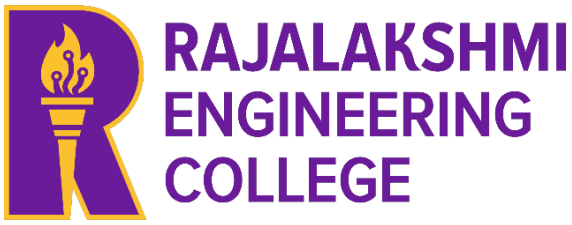
CB23F35 –

LOW-CODE APPLICATION DEVELOPMENT USING ZOHO CREATOR

in

BACHELOR OF ENGINEERING

COMPUTER SCIENCE AND ENGINEERING



RAJALAKSHMI ENGINEERING COLLEGE

(An Autonomous Institution, Affiliated to Anna University, Chennai)

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RAJALAKSHMI ENGINEERING COLLEGE

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BONAFIDE CERTIFICATE

Certified that this project report titled "Hostel Room Issue Tracker" is the bonafide work of M. Siva Swathi (Roll No. 240701516), who carried out the individual project work on the Zoho Creator low-code platform for the course CB23F35 – Low-Code Application Development using Zoho Creator, during the academic year 2026–2027, Semester V, under my supervision.

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INTERNAL EXAMINER	**EXTERNAL EXAMINER**

DECLARATION

I, M. Siva Swathi (Roll No. 240701516), hereby declare that the project report titled "Hostel Room Issue Tracker", submitted for the End Semester Examination of the course CB23F35 – Low-Code Application Development using Zoho Creator, is a record of the individual project work carried out by me.

The proposed application is designed to manage hostel room maintenance complaints submitted by students. The report describes the data model, forms, reports, workflows, Deluge-based automation where configured, dashboard, Blueprint process, approval process, role-based access and notification schedule planned for the application.

I further declare that this work has not been submitted, in part or in full, for the award of any other degree or course credit. Any external documentation or resources used have been acknowledged in the references.

Place: Chennai

Date: _____

SIGNATURE OF THE STUDENT

M. SIVA SWATHI

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I also acknowledge Zoho Corporation for providing the Zoho Creator platform and its learning resources. Finally, I thank my family and friends for their encouragement and support.

M. SIVA SWATHI

ABSTRACT

Hostel maintenance complaints are often communicated through verbal messages, phone calls, informal messaging groups or paper registers. When complaints are handled through separate channels, students may not know whether their issues have been recorded, wardens may find it difficult to monitor pending work, and maintenance staff may not receive clear assignment details. Complaint images, status updates and resolution records can also become scattered.

Hostel Room Issue Tracker is a centralised hostel maintenance complaint management application designed using Zoho Creator, a low-code application development platform. The application is intended to provide a structured process for students to register room-related issues, attach supporting images and track complaint progress. The data model contains six forms: Student Registration, Room Details, Issue Category, Complaint Registration, Maintenance Staff Details and Resolution Update. Lookup relationships connect students, rooms, issue categories, complaints and maintenance assignments.

The proposed process begins when a student submits a complaint. The complaint is recorded with a unique Complaint ID and an initial status. A warden reviews the complaint and approves or rejects it. Approved complaints can be assigned to maintenance staff, who update work progress and submit resolution details. Reports provide views of all, pending and resolved complaints, staff assignments and complaint dates. A dashboard is planned to display complaint KPIs, category and status charts, recent complaints and staff performance. A Blueprint defines the complaint lifecycle, while an approval process, role-based permissions, email notifications and a daily pending-complaint reminder support process control.

The application aims to reduce dependence on manual follow-up, improve complaint visibility and maintain a consistent record from submission to closure. Actual deployment status, screenshots, test outcomes, portal availability and plan-dependent features must be verified in the live Zoho Creator application before final submission.

Keywords: Zoho Creator; Low-Code Development; Hostel Management; Complaint Tracking; Deluge; Workflow Automation; Blueprint; Role-Based Access Control.

PROJECT AT A GLANCE

Attribute	Details
Project Title	Hostel Room Issue Tracker
Application Domain	Hostel Administration – room maintenance complaint registration and tracking
Student Name & Roll No.	M. Siva Swathi – 240701516
Course	CB23F35 – Low-Code Application Development using Zoho Creator
Platform	Zoho Creator (edition to be confirmed)
Application Name	Hostel Room Issue Tracker
Number of Forms	6 forms: Student Registration, Room Details, Issue Category, Complaint Registration, Maintenance Staff Details and Resolution Update
Reports	5 planned reports: All Complaints, Pending Complaints, Resolved Complaints, Staff Assignment Report and Complaint Calendar
Dashboard	Hostel Issue Dashboard with KPIs, charts, recent complaints and staff performance summary
Workflow Components	5 planned form/workflow automations, Blueprint, approval process and daily schedule
Primary Roles	Student, Warden, Maintenance Staff and Admin
Integrations	Email notifications using supported Zoho Creator email actions or integrations
Deployment	Development/trial application; live URL and deployment status to be verified
Evidence Note	This report is a project-report draft based on the supplied requirements. Mark implementation and test status only after checking the live application.

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LIST OF ABBREVIATIONS

Abbreviation	Expansion
API	Application Programming Interface
ER	Entity–Relationship
ID	Identification
IST	Indian Standard Time
LCAP	Low-Code Application Platform
RBAC	Role-Based Access Control
UAT	User Acceptance Testing
UI	User Interface
URL	Uniform Resource Locator
WF	Workflow
Zia	Zoho's AI assistant

CHAPTER 1

INTRODUCTION

1.1 Background

Hostel facilities require regular maintenance to provide students with a safe and comfortable living environment. Common issues include electrical faults, plumbing leaks, damaged furniture, cleaning requirements and internet connectivity problems. Students need a simple way to report these issues, while wardens and maintenance teams need a reliable method to review, prioritise, assign and resolve them.

In a manual process, students may contact a warden in person or send a message. The warden may forward the issue to a maintenance worker and later ask for an update. Complaint details, photographs, assignment information and completion dates may remain in different messages or registers. This makes it difficult to identify overdue complaints, monitor staff workload and provide students with a clear status.

Hostel Room Issue Tracker proposes a centralised application using Zoho Creator. The application stores student and room information, classifies complaints, records images, supports approval and assignment, and maintains resolution details. It is intended to provide a single source of information for students, wardens, maintenance staff and administrators.

1.2 Problem Statement

The project addresses the absence of a structured and transparent process for recording and managing hostel room maintenance complaints. Manual communication can lead to missing complaint details, delayed assignment, unclear responsibility and limited status visibility.

- Students may not receive a unique reference number or confirmation after reporting an issue.
- Wardens may find it difficult to monitor all pending complaints and identify urgent issues.
- Maintenance staff may receive incomplete details about the location, category or priority of a complaint.

- Photographs and resolution evidence may not be stored with the complaint record.
- Students may not know whether a complaint is submitted, approved, assigned, in progress or resolved.
- Management may lack consolidated reports about complaint categories, status and staff workload.
- Shared registers or message groups provide limited record-level access control and auditability.

1.3 Objectives

1. Develop a centralised Zoho Creator application for hostel room maintenance complaints.
2. Create related forms for students, rooms, issue categories, complaints, maintenance staff and resolution updates.
3. Provide unique complaint identification and structured status tracking.
4. Enable warden review and approval or rejection of submitted complaints.
5. Allow approved complaints to be assigned to suitable maintenance staff.
6. Record work progress, completion information, resolution images and student feedback.
7. Generate reports and a dashboard for complaint monitoring and staff performance.
8. Configure role-based access and email notifications to improve communication.
9. Provide a daily reminder for complaints that remain unresolved.

1.4 Scope of the Project

The project scope includes student registration, hostel room master data, issue category management, complaint submission with image upload, warden approval, maintenance staff assignment, work-progress updates, resolution records, reports, dashboard visualisation, Blueprint transitions, role-based access and email reminders.

The proposed application contains six forms, five reports, one dashboard, workflow automations, one complaint Blueprint, one warden approval process and one daily

pending-complaint schedule. Portal access is included subject to the capabilities and limits of the Zoho Creator edition used.

Features such as SMS or WhatsApp notifications, predictive maintenance, IoT-based fault detection, integration with external hostel ERP systems and payment processing are outside the current scope and may be considered as future enhancements.

1.5 Justification for a Low-Code Approach

Zoho Creator is selected because the project requires relational forms, lookup fields, reports, dashboards, workflow automation, approval routing and role-based permissions. These capabilities can be configured through a visual development environment. Deluge can be used for additional validation, record retrieval, status updates and email actions where built-in configuration is insufficient. This reduces the amount of separate frontend, backend, database and hosting code required for a prototype.

CHAPTER 2

LITERATURE SURVEY AND SYSTEM STUDY

2.1 Study of the Existing System

The existing hostel complaint process is commonly handled through direct conversations, telephone calls, messaging applications or written registers. A student reports a fault to a warden. The warden identifies the relevant maintenance person and forwards the information. The staff member visits the room, performs the work and informs the warden. The student may or may not receive a formal completion update.

1. Student notices a maintenance issue and informs the warden.
2. Warden records the issue manually or forwards a message to maintenance staff.
3. Maintenance staff receives the location and issue details.
4. Staff visits the room and performs the repair or service.
5. Warden or staff communicates completion informally to the student.
6. Records are retained in messages, notebooks or spreadsheets, if recorded at all.

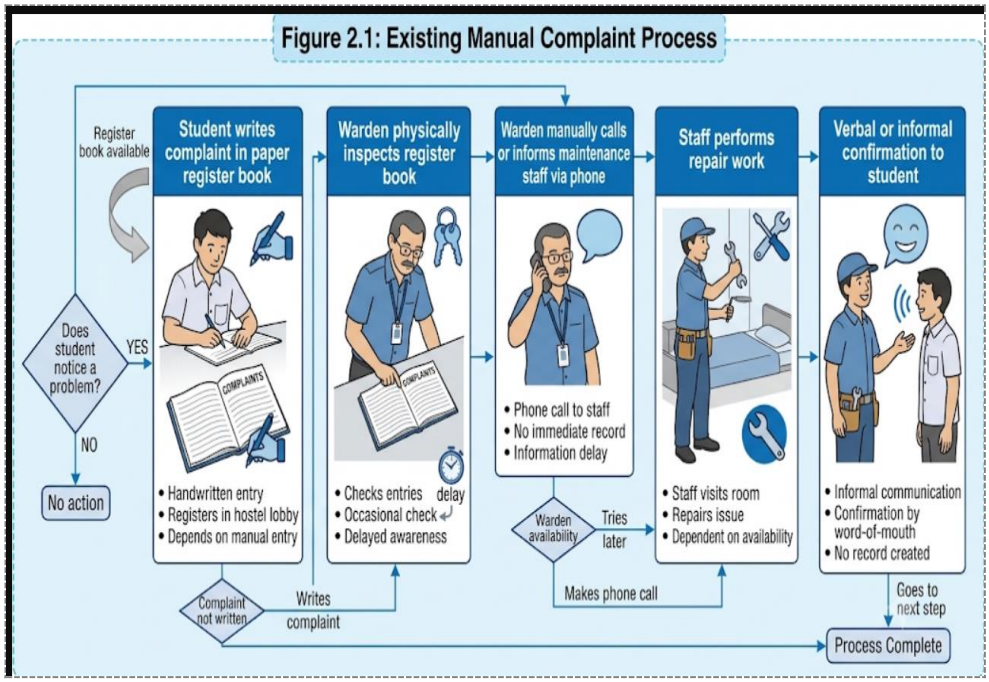


Figure 2.1: Existing Manual Complaint Process

2.2 Limitations of the Existing System

- No central complaint register with a consistent unique identifier.
- Complaint details and images may be distributed across communication channels.
- Manual forwarding can delay assignment and action.
- No standard status lifecycle for students to track progress.
- Limited visibility of urgent, pending and overdue complaints.
- No structured record of staff assignment, work start and completion dates.
- Difficult to prepare category-wise, status-wise and staff-wise summaries.
- Limited control over who can view student and complaint information.

2.3 Low-Code versus Traditional Development

Table 2.1: Low-Code and Traditional Development Comparison

Criterion	Traditional Development	Zoho Creator Low-Code
Development effort	Separate frontend, backend, database and hosting components are developed.	Forms, reports, pages, workflows and permissions are configured on one platform.
Skills	Requires programming and infrastructure knowledge across multiple layers.	Requires platform configuration and Deluge knowledge.
Data management	Database schema and APIs must be created separately.	Forms and lookup fields provide structured data capture.
Automation	Backend jobs and notification services must be implemented.	Built-in workflows, schedules and approval features can be configured.
Maintenance	Code changes and deployment may be required.	Many changes can be made through Creator configuration.
Customisation	Offers broad architectural freedom.	Customisation is bounded by platform features and scripting.
Cost	Depends on development, hosting and maintenance resources.	Depends on Creator edition, users and feature availability.

2.4 Literature Survey

The project is informed by platform documentation and low-code development references. The following resources are relevant to the design and implementation. Exact feature availability should be checked against the edition used.

Table 2.2: Literature Survey

No.	Source / Resource	Relevance
1	Zoho Creator Help Documentation	Forms, lookup fields, reports, pages, workflows and application permissions.
2	Deluge Script Reference	Record retrieval, conditions, validation, updates and email actions.
3	Zoho Creator Blueprint and Approval documentation	State-based process management and approval routing.
4	Zoho Creator Portal and User Management documentation	External user access, roles and data visibility controls.
5	Zoho Creator Schedules and Notifications documentation	Recurring checks and automated reminders.

2.5 Emerging Trends in Low-Code Development

Low-code platforms support rapid application development through reusable components, visual data modelling, cloud deployment and workflow automation. Current development trends include AI-assisted application creation, integration with business services, mobile-first access, analytics and citizen development. In this project, the main low-code concepts are represented through related forms, reports, a dashboard, workflow automation, Blueprint, approval routing and permission configuration. Zia-assisted generation may be used during application creation, but the final application should be checked and tested by the developer.

2.6 Proposed System

The proposed system centralises hostel maintenance complaint information in Zoho Creator. Students are registered and associated with hostel room records. A student submits a complaint by selecting a room and issue category, entering a title and description, attaching an image if available and choosing a priority. The record receives a Complaint ID and enters the complaint lifecycle.The warden reviews the complaint and approves or rejects it. Approved complaints proceed to staff assignment. Maintenance staff update the work status and enter resolution details.

CHAPTER 3

REQUIREMENT ANALYSIS

3.1 Stakeholders and User Roles

Table 3.1: Stakeholders and User Roles

Role	Description	Responsibilities
Student	Hostel resident who reports maintenance issues.	Submit complaints, upload images, view own complaints and track resolution.
Warden	Hostel authority responsible for complaint review.	View complaints, approve/reject submissions and assign maintenance staff.
Maintenance Staff	Person responsible for repair or service work.	View assigned complaints, update work progress and submit resolution details.
Admin	Application administrator.	Manage master data, users, permissions, configuration and overall application records.

3.2 Functional Requirements

Table 3.2: Functional Requirements

ID	Requirement	Role	Priority
FR-01	Register student details with a unique register number and contact information.	Admin	High
FR-02	Maintain hostel, block, floor, room number and room capacity information.	Admin	High
FR-03	Maintain issue categories, descriptions and default priorities.	Admin	High
FR-04	Allow a student to submit a complaint with room, category, title, description, priority and optional image.	Student	High
FR-05	Generate a unique Complaint ID and initialise complaint status.	System	High
FR-06	Allow the warden to review and approve or reject complaints with an appropriate reason.	Warden	High
FR-07	Allow approved complaints to be assigned to maintenance staff.	Warden	High
FR-08	Allow assigned staff to update work progress and resolution details.	Maintenance Staff	High

ID	Requirement	Role	Priority
FR-09	Store assignment, work-start and completion dates and resolution evidence.	Maintenance Staff	High
FR-10	Provide all, pending, resolved, staff assignment and calendar reports.	Warden/Admin	High
FR-11	Display complaint KPIs, category and status charts and recent records.	Warden/Admin	Medium
FR-12	Send notifications to the warden, student and assigned staff at relevant stages.	System	High
FR-13	Send a daily reminder for complaints that are not resolved, rejected or closed.	System	Medium
FR-14	Restrict students to their own complaint records and expose only permitted actions.	System	High
FR-15	Maintain complaint lifecycle transitions through Blueprint where available.	Warden/Staff/Admin	High

3.3 Non-Functional Requirements

Table 3.3: Non-Functional Requirements

ID	Category	Requirement
NFR-01	Usability	Forms and reports should use clear labels, logical grouping and a consistent blue-and-white theme.
NFR-02	Security	Users should access only records, fields and actions permitted by their roles.
NFR-03	Data Integrity	Required fields, unique identifiers and lookup relationships should prevent incomplete or inconsistent records.
NFR-04	Availability	The application should be accessible through the supported Zoho Creator web interface.
NFR-05	Maintainability	Administrators should be able to update room details, categories and staff records.
NFR-06	Performance	Reports and dashboards should load within acceptable time for the expected hostel dataset.
NFR-07	Auditability	Status changes, assignment details and resolution information should be traceable.
NFR-08	Scalability	The data model should support additional hostels, blocks, rooms and complaint records.

3.4 Hardware and Software Requirements

Table 3.4: Hardware and Software Requirements

Category	Specification
Platform	Zoho Creator (edition / trial status to be confirmed)
Configuration / scripting	Zoho Creator visual builder and Deluge
Browser	Google Chrome, Microsoft Edge or Mozilla Firefox
Operating system	Windows 10/11 or another supported desktop operating system
Client hardware	Intel Core i3 or equivalent, 8 GB RAM recommended, stable internet connection
Additional services	Zoho Creator email actions or supported Zoho email integration
Mobile access	Zoho Creator mobile application or configured portal, subject to plan availability

3.5 Use Case Model

The main use cases are Submit Complaint, Review Complaint, Approve or Reject Complaint, Assign Maintenance Staff, Update Work Progress, Submit Resolution, Track Complaint and View Reports. The primary actors are Student, Warden, Maintenance Staff and Admin.

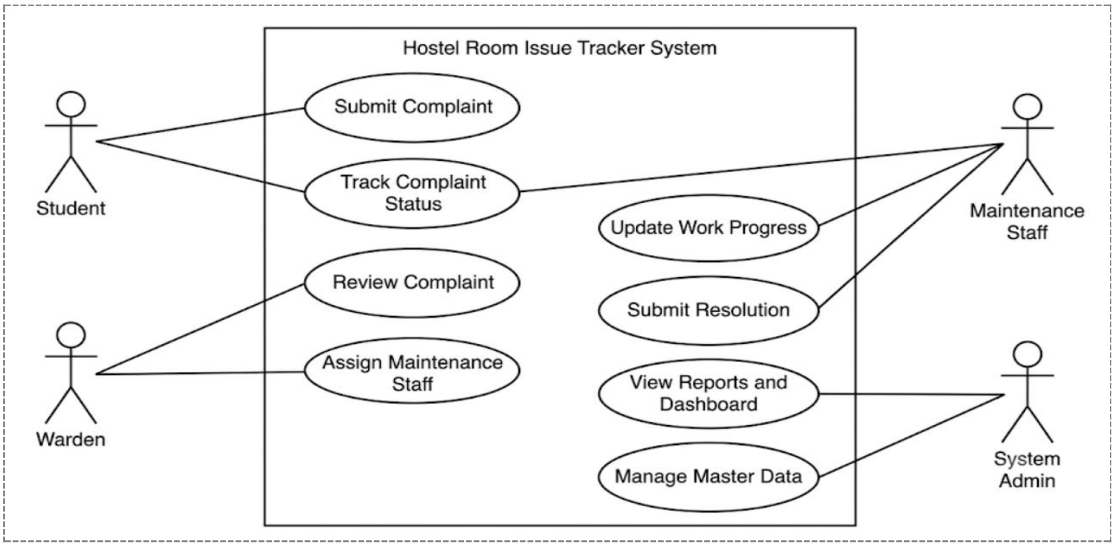


Figure 3.1: Use Case Model

Table 3.5: Use Case Description

Use Case	Actor	Main Flow	Post-condition
UC-01 Submit Complaint	Student	Open complaint form → select room and category → enter title and description → attach image if required → submit.	Complaint record is stored with a unique ID and initial status.
UC-02 Review Complaint	Warden	Open pending complaints → inspect issue details and image → approve or reject → enter reason if required.	Complaint status is updated and the student is notified.
UC-03 Assign Staff	Warden	Open approved complaint → select available staff member → set assignment date → confirm assignment.	Complaint is assigned and staff notification is initiated.
UC-04 Update Work	Maintenance Staff	Open assigned complaint → start work → update progress → enter work description and dates.	Work progress is recorded.
UC-05 Submit Resolution	Maintenance Staff	Select complaint → enter completed date and resolution details → attach evidence → submit.	Resolution record is linked to the complaint and status can move to Resolved.
UC-06 Track Complaint	Student	Open own complaint list → select complaint → review status and permitted resolution information.	Student can see the latest permitted complaint information.

CHAPTER 4

SYSTEM DESIGN

4.1 System Architecture

The system is organised into four logical layers. The presentation layer contains the user interfaces available to students, wardens, maintenance staff and administrators. The application layer contains the six forms, reports, dashboard, workflows, Blueprint and approval configuration. The data layer stores student, room, category, complaint, staff and resolution records. The notification layer uses supported Zoho Creator email actions and the scheduled reminder.

Table 4.1: System Architecture Layers

Layer	Components	Purpose
Presentation	Student, Warden, Maintenance Staff and Admin interfaces	Capture information and display permitted records.
Application	Six forms, five reports, dashboard, workflows, Blueprint and approval	Apply complaint-management business rules.
Data	Student, Room, Issue Category, Complaint, Staff and Resolution records	Store related operational data.
Notification	Email actions and daily schedule	Notify users about submission, decisions, assignment, resolution and pending work.

Logical flow: Users → Zoho Creator UI → Forms and business rules → Creator data store → Reports / Dashboard; workflow events and the daily schedule → Email notifications.

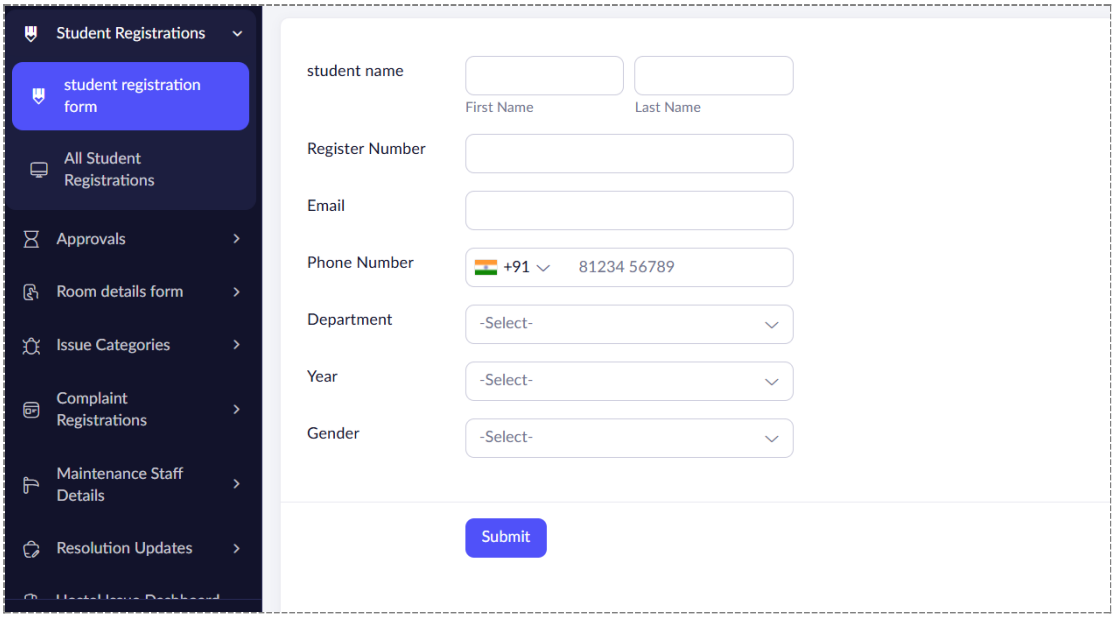


Figure 4.1: System Architecture

4.2 Data Model

The data model contains six entities. Complaint Registration is the central transaction form. It references Student Registration, Room Details and Issue Category. Resolution Update references Complaint Registration and Maintenance Staff Details. This structure reduces repeated entry and supports report generation.

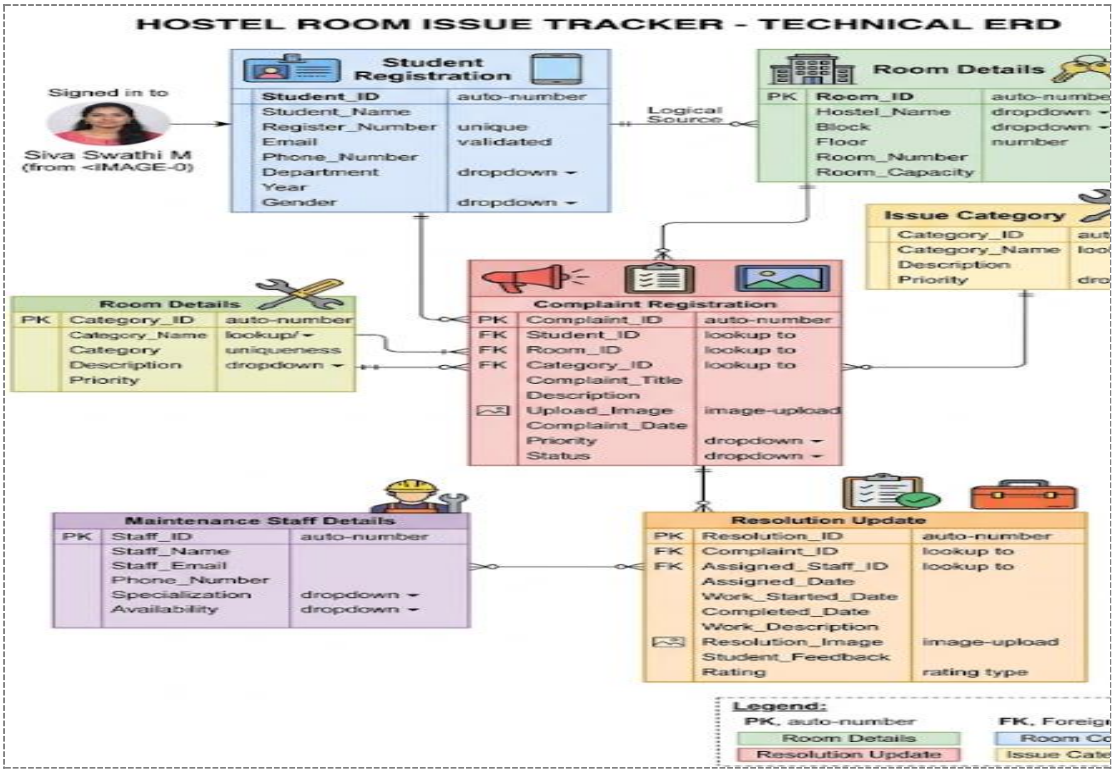


Figure 4.2: Entity Relationship Model

Table 4.2: Entities and Relationships

Entity	Key Fields	Relationship
Student Registration	Student ID, Student Name, Register Number, Email, Phone, Department, Year, Gender	One student can submit many complaints.
Room Details	Room ID, Hostel Name, Block, Floor, Room Number, Room Capacity	One room can be associated with many complaints.
Issue Category	Category ID, Category Name, Description, Priority	One category can classify many complaints.
Complaint Registration	Complaint ID, Student, Room, Issue Category, Title, Description, Image, Date, Priority, Status	Central complaint record; links student, room and category.
Maintenance Staff Details	Staff ID, Staff Name, Staff Email, Phone, Specialization, Availability	One staff member can be assigned to multiple complaints over time.
Resolution Update	Resolution ID, Complaint, Assigned Staff, Assigned Date, Work Started Date, Completed Date, Work Description, Image, Feedback, Rating	Stores assignment and resolution information for a complaint.

4.3 Form Design

4.3.1 Student Registration Form

Table 4.3: Field Specification – Student Registration

Field	Field Type	Validation / Purpose
Student ID	Auto Number	Unique system-generated identifier.
Student Name	Single Line	Required.
Register Number	Single Line	Required and unique.
Email	Email	Required; valid email format.
Phone Number	Phone	Validate supported phone format.
Department	Dropdown	Department selection.
Year	Dropdown	1st, 2nd, 3rd or 4th Year.
Gender	Dropdown	Configured options.

4.3.2 Room Details Form

Table 4.4: Field Specification – Room Details

Field	Field Type	Validation / Purpose
Room ID	Auto Number	Unique system-generated identifier.
Hostel Name	Dropdown	Required hostel selection.
Block	Dropdown	Required block selection.
Floor	Number	Floor number.
Room Number	Single Line	Required room identifier.
Room Capacity	Number	Room occupancy capacity.

4.3.3 Issue Category Form

Table 4.5: Field Specification – Issue Category

Field	Field Type	Validation / Purpose
Category ID	Auto Number	Unique identifier.
Category Name	Dropdown	Electrical, Plumbing, Furniture, Cleaning, Internet, Other.
Description	Multi Line	Optional category description.
Priority	Dropdown	Low, Medium, High, Emergency.

4.3.4 Complaint Registration Form

Table 4.6: Field Specification – Complaint Registration

Field	Field Type	Validation / Purpose
Complaint ID	Auto Number	Unique complaint reference.
Student	Lookup	Lookup to Student Registration; required.
Room	Lookup	Lookup to Room Details; required.
Issue Category	Lookup	Lookup to Issue Category; required.
Complaint Title	Single Line	Required.
Description	Multi Line	Required detailed issue description.
Upload Image	Image Upload	Optional supporting image.
Complaint Date	Date	Date of complaint submission.
Priority	Dropdown	Low, Medium, High, Emergency; required.
Status	Dropdown	Submitted, Pending Approval, Approved, Rejected, Assigned, In Progress, Resolved, Closed; default Submitted.

4.3.5 Maintenance Staff Details Form

Table 4.7: Field Specification – Maintenance Staff Details

Field	Field Type	Validation / Purpose
Staff ID	Auto Number	Unique staff identifier.
Staff Name	Single Line	Required.
Staff Email	Email	Required; valid email.
Phone Number	Phone	Contact information.
Specialization	Dropdown	Electrical, Plumbing, Furniture, Cleaning, Internet, General Maintenance.
Availability	Dropdown	Available, Busy, Unavailable.

4.3.6 Resolution Update Form

Table 4.8: Field Specification – Resolution Update

Field	Field Type	Validation / Purpose
Resolution ID	Auto Number	Unique resolution identifier.
Complaint	Lookup	Lookup to Complaint Registration; required.
Assigned Staff	Lookup	Lookup to Maintenance Staff Details; required.
Assigned Date	Date	Date of assignment.

Field	Field Type	Validation / Purpose
Work Started Date	Date	Date work began.
Completed Date	Date	Date work was completed.
Work Description	Multi Line	Work performed and resolution details.
Resolution Image	Image Upload	Optional evidence of completed work.
Student Feedback	Multi Line	Feedback after resolution.
Rating	Rating	Student rating, where enabled.

4.4 Process and Workflow Design

The intended process begins with student and room master records. The student submits a complaint and the application stores it with Submitted status. The warden reviews it and moves it to Pending Approval before making an approval decision. Approved complaints are assigned to a maintenance staff member. The staff member starts work, updates progress and submits a resolution record. The complaint can then be marked Resolved and later Closed after the required confirmation.

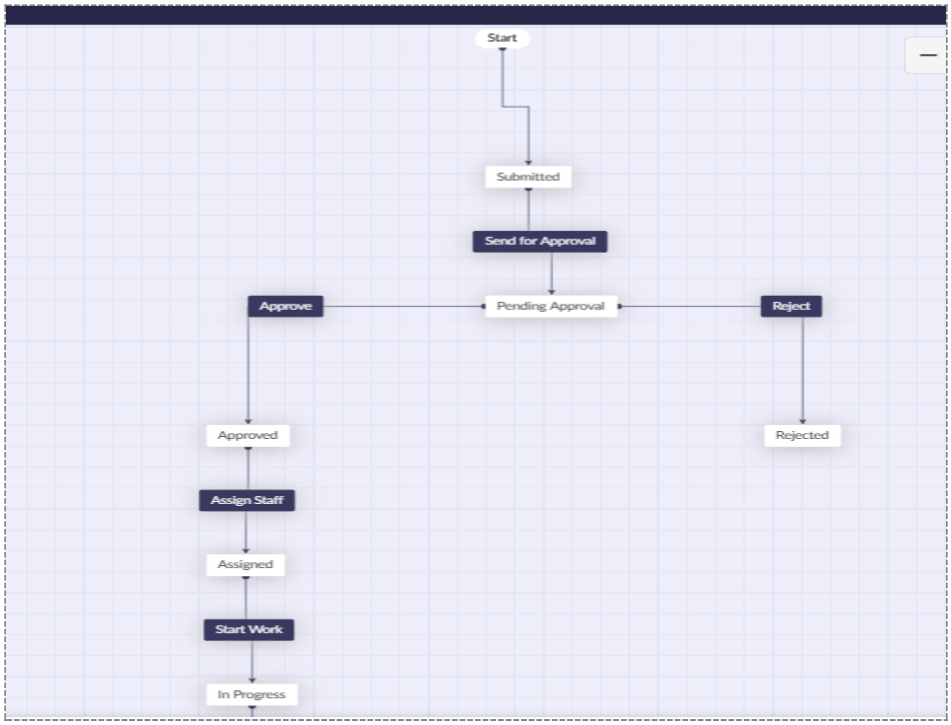


Figure 4.3: Complaint Approval and Resolution Flow

Table 4.9: Complaint Process Stages

Stage	Responsible Role	Action	Next Status
Submission	Student	Enter complaint details and submit.	Submitted
Review queue	Warden	Review complaint and route for decision.	Pending Approval
Decision	Warden	Approve or reject with reason where applicable.	Approved / Rejected
Assignment	Warden	Select staff member and record assignment date.	Assigned
Work	Maintenance Staff	Start work and update progress.	In Progress
Resolution	Maintenance Staff	Submit work description and completion details.	Resolved
Closure	Warden / Admin	Verify resolution and close complaint.	Closed

4.5 User Interface and Navigation Design

The application is intended to use a modern blue-and-white theme with clear labels and a simple navigation menu. Navigation should expose only the forms, reports and dashboard relevant to the signed-in role. Students should see complaint submission and their own complaint tracking views. Wardens should see complaint review, assignment and reports. Maintenance staff should see assigned work and resolution entry. Administrators should manage master data and application configuration.

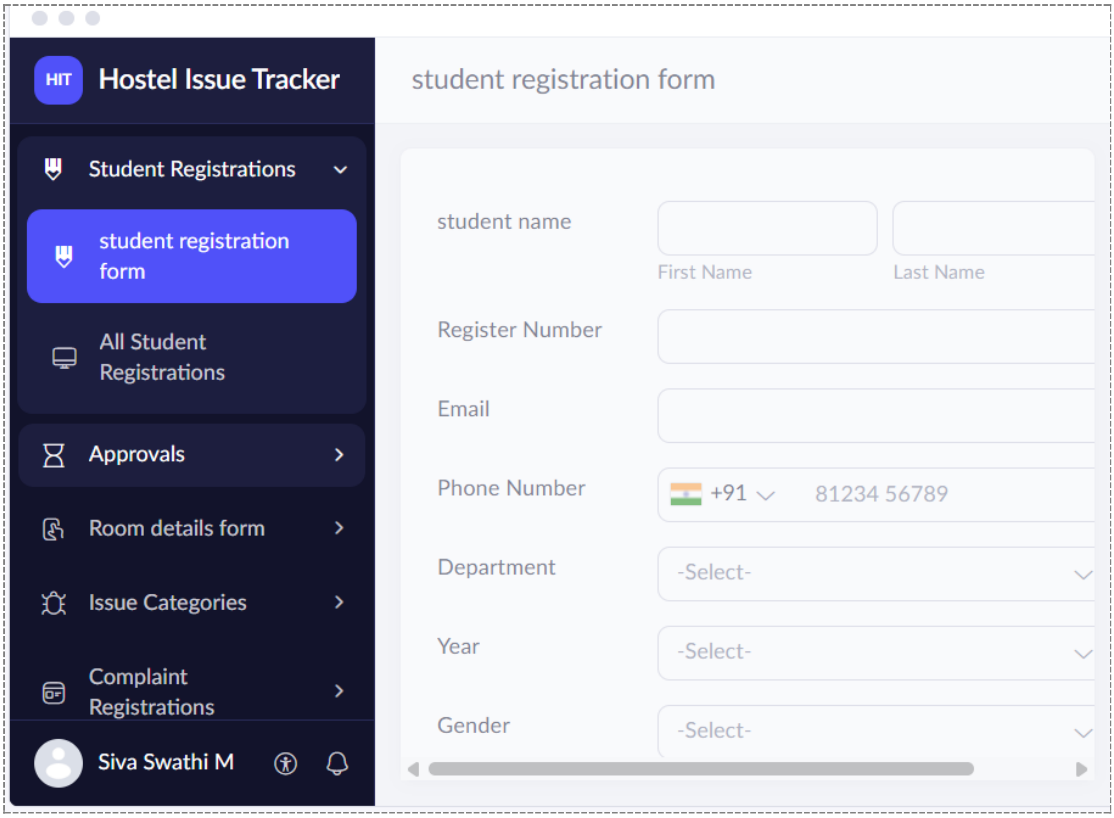


Figure 4.4: Application Navigation Structure

4.6 Access Control Design

Table 4.10: Role-Permission Matrix

Role	Permitted Access	Restrictions
Student	Submit complaints; view own complaints; track status; view permitted resolution details.	Must not view or edit other students' private complaints or approve requests.
Warden	View all complaints; approve/reject; assign staff; view reports and dashboard.	Administrative configuration restricted unless also assigned Admin role.
Maintenance Staff	View assigned complaints; update work progress; submit resolution details.	Should not approve complaints or access unrelated student records.
Admin	Full application administration and data management.	Access should be assigned only to authorised administrators.

CHAPTER 5

IMPLEMENTATION

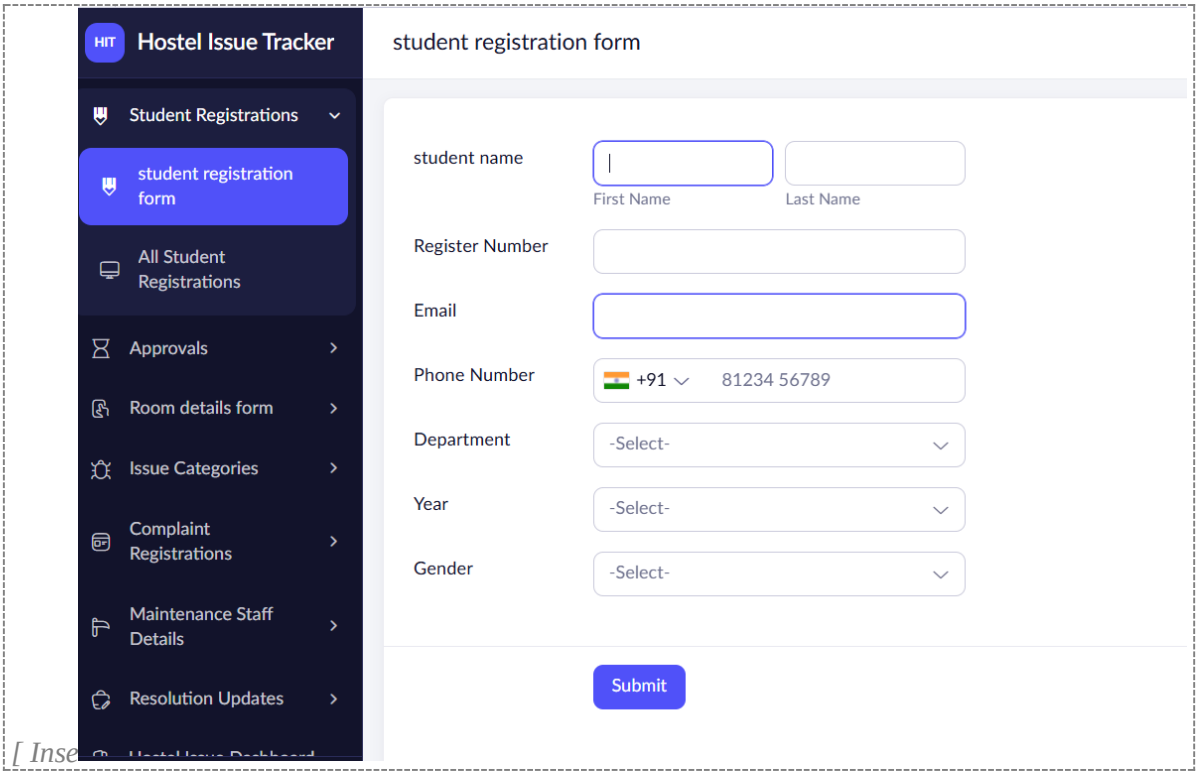
5.1 Application Setup

The application is created in Zoho Creator with the name “Hostel Room Issue Tracker”. Zia may be used to generate the initial application structure. The generated forms, fields, lookup relationships and navigation must be reviewed manually. The application should use the blue-and-white theme and avoid duplicate forms or unnecessary modules.

Before final submission, record the exact Creator edition, application link name, live URL, creation date and deployment status from the account. If a feature is unavailable in the trial edition, document it as a limitation and provide the manual configuration attempted.

5.2 Forms and Data Capture

The six forms described in Section 4.3 form the data-capture foundation. Auto Number fields provide identifiers. Lookup fields connect related forms. Required-field validation should be enabled for important data. The Complaint Registration form is the main transaction form and should be checked to ensure that Student, Room and Issue Category lookups display the correct source records.



[Inse

Figure 5.1: Student Registration Form

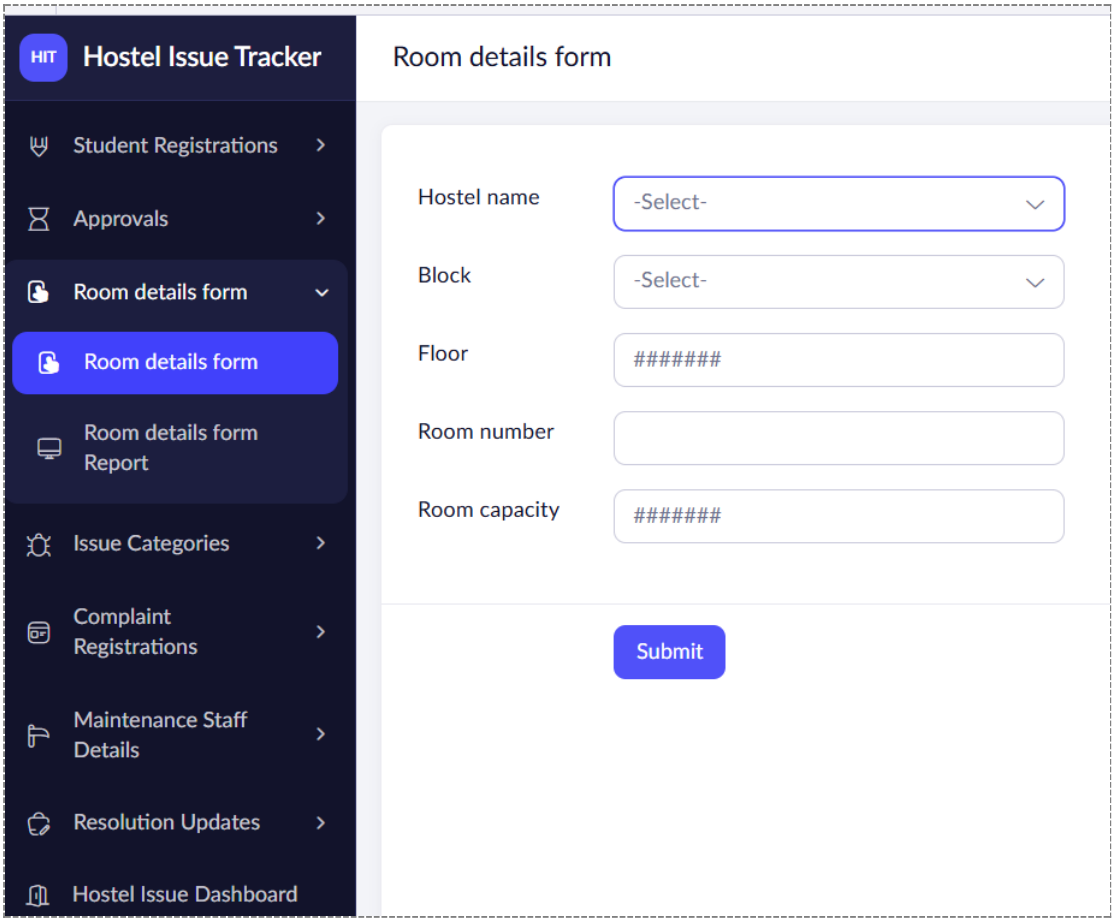


Figure 5.2: Room Details Form

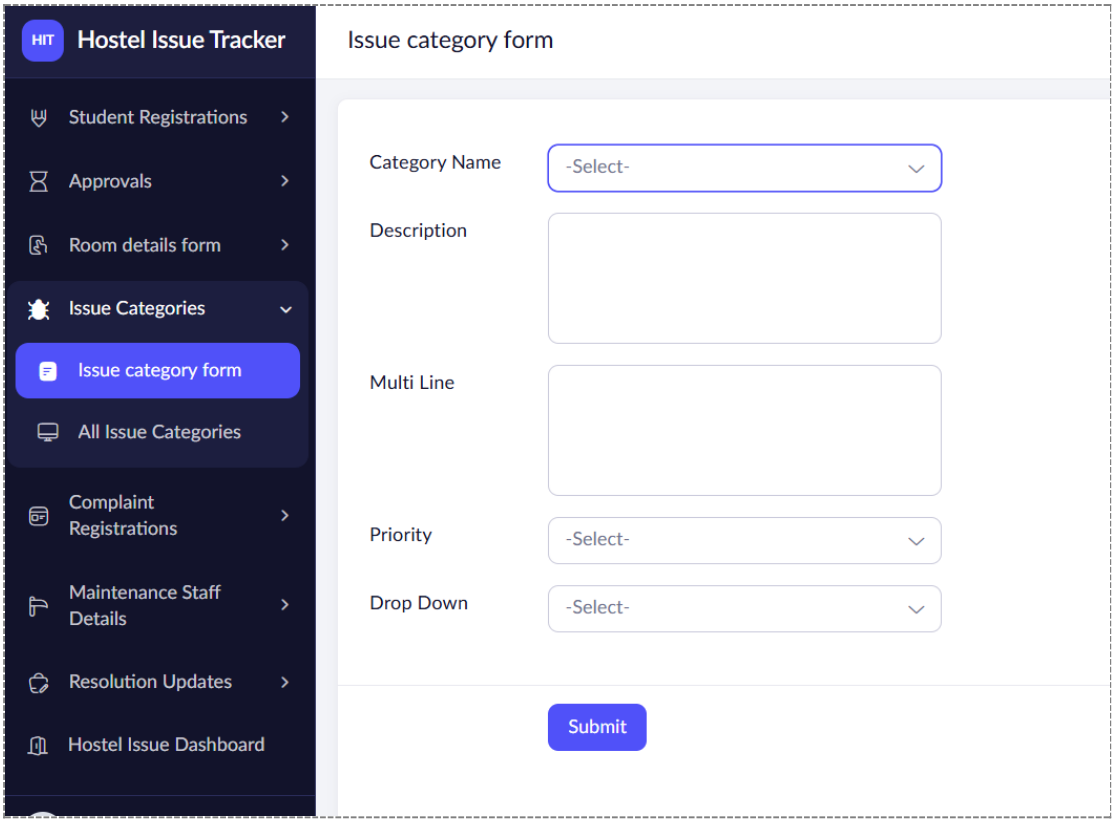


Figure 5.3: Issue Category Form

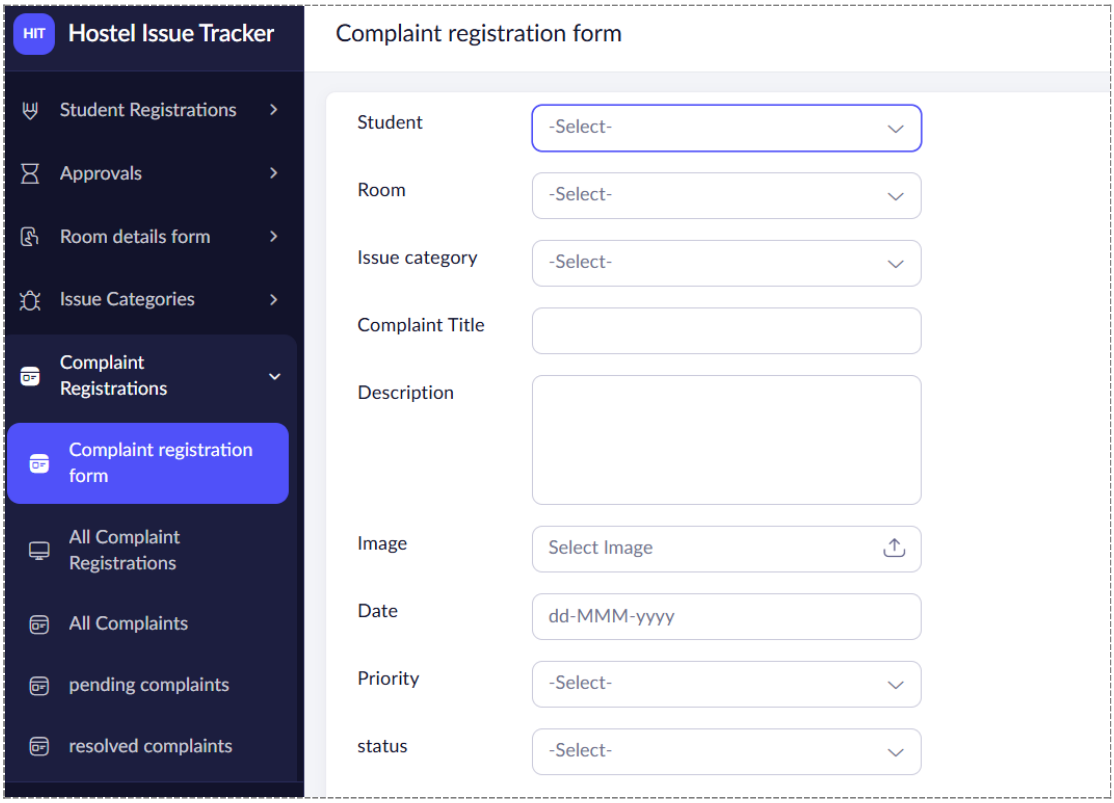


Figure 5.4: Complaint Registration Form

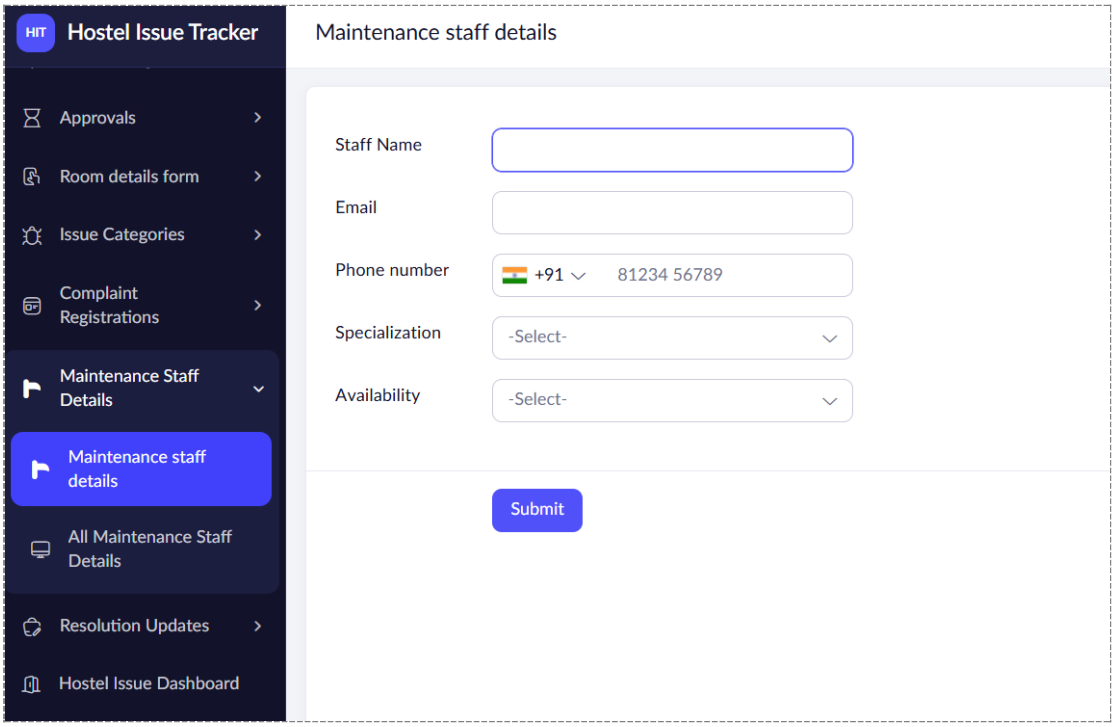


Figure 5.5: Maintenance Staff Details Form

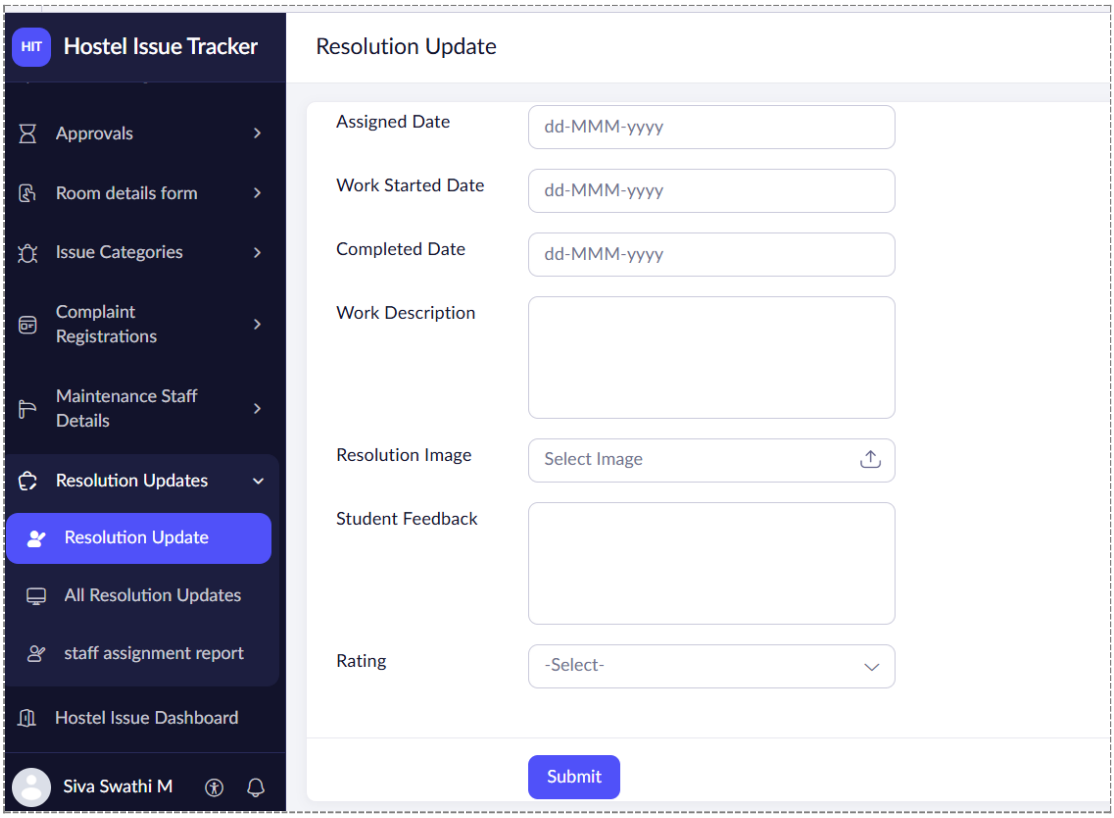


Figure 5.6: Resolution Update Form

5.3 Reports

Table 5.1: Reports Implemented / Planned

No.	Report Name	Source / Content	Filter or Layout
1	All Complaints	Complaint ID, Student, Room, Category, Priority and Status.	All complaint records; sortable list.
2	Pending Complaints	Complaints awaiting warden decision.	Status = Pending Approval.
3	Resolved Complaints	Complaints marked resolved.	Status = Resolved.
4	Staff Assignment Report	Complaint ID, Assigned Staff, Assigned Date, Work Started Date and Completed Date.	Based on Resolution Update and linked complaint data.
5	Complaint Calendar	Complaint records displayed by Complaint Date.	Calendar view using Complaint Date.

5.4 Pages and Dashboard

The planned page is named “Hostel Issue Dashboard”. It contains four KPI panels: Total Complaints, Pending Complaints, Approved Complaints and Resolved Complaints. A pie chart groups complaints by category, while a bar chart groups complaints by status. A recent complaints report provides the latest records. A staff performance summary can show assignment and completion counts based on available resolution data. All dashboard components should use live application data and be checked after new records are added.

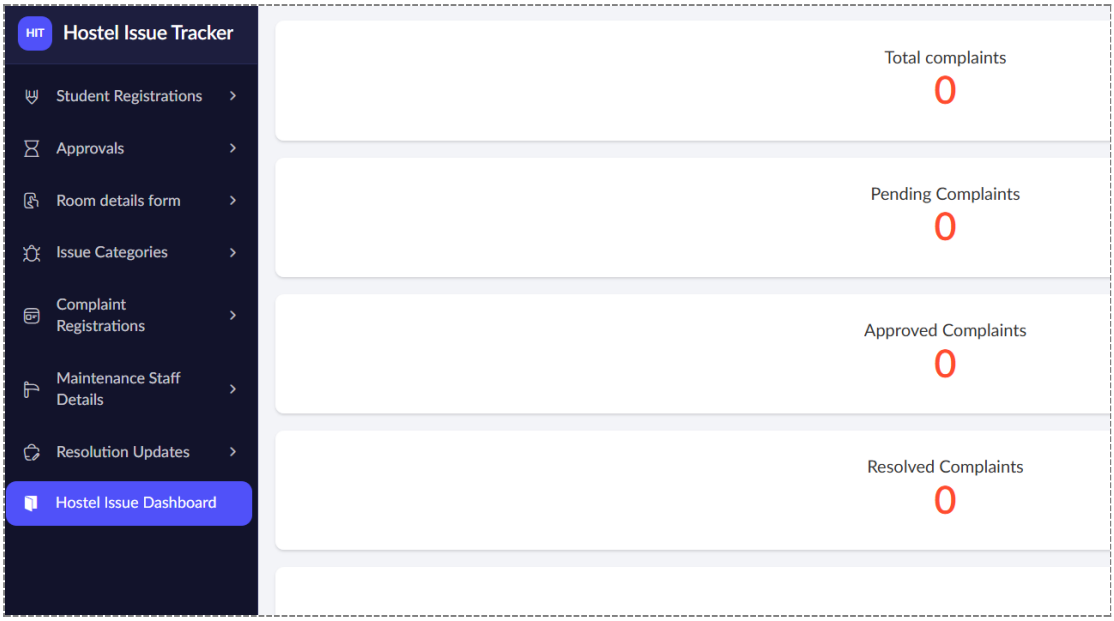


Figure 5.7: Hostel Issue Dashboard

5.5 Deluge Scripting

Deluge can be used when the built-in form, workflow or report configuration does not fully implement a business rule. The following are representative logic patterns. Link names must be copied from the actual Zoho Creator forms and fields before scripts are used. These samples are not guaranteed to run without adapting the link names and event context.

5.5.1 Complaint Submission Validation – Logic Outline

On successful complaint submission, allow Creator's Auto Number field to generate the Complaint ID. Set the initial Status field to Submitted using the form's default value or a supported workflow action. Do not insert a second complaint record from the workflow; operate on the current record to avoid duplicates.

5.5.2 Staff Assignment – Logic Outline

When the warden assigns an approved complaint, validate that a staff member is selected and that the staff availability is appropriate. Create or update the linked Resolution Update record according to the application's chosen design. Update the complaint status to Assigned and notify the selected staff member. Prevent duplicate assignment records for the same complaint unless reassignment is explicitly supported.

5.5.3 Resolution Submission – Logic Outline

When a resolution record is submitted, validate the linked complaint and assigned staff. Check that the completed date is not earlier than the work-start date. Update the linked complaint status to Resolved only after the resolution record is successfully saved and the required fields are present. Avoid creating duplicate resolution records.

5.6 Workflows and Automation

Table 5.2: Workflow and Automation Summary

Workflow	Trigger	Actions
Complaint Submission Workflow	Complaint Registration successfully submitted.	Use generated Complaint ID; initialise Submitted status; ensure record appears in All Complaints; do not create duplicate records.
Warden Notification Workflow	New complaint submitted.	Send complaint ID, student, room, category, priority and description to configured warden email.
Approval Notification Workflow	Complaint routed for warden approval and decision recorded.	Notify warden for review; do not auto-approve; notify student after approval or rejection.
Staff Assignment Workflow	Approved complaint assigned to staff.	Set status to Assigned; email assigned staff with complaint, room, category and priority.
Resolution Notification Workflow	Resolution Update successfully submitted.	Update linked complaint to Resolved when valid; notify student with complaint, room, resolution and completion date.

5.7 Email Integration

Email actions should use configured and verified recipient addresses. Dynamic form values should be used for complaint and student details. The application should test each notification before enabling it for regular use. The warden email address should be stored in a configuration location or entered through an approved application setting rather than being scattered as hardcoded values across scripts.

Table 5.3: Email Notification Events

Event	Recipient	Information
New complaint submitted	Warden	Complaint ID, student name, room, category, priority and description.

Event	Recipient	Information
Complaint sent for approval	Warden	Complaint ID, student, room and issue description.
Approved or rejected	Student	Decision, complaint ID and rejection reason if applicable.
Staff assigned	Assigned Staff	Complaint ID, room, issue category, priority and work instructions.
Complaint resolved	Student	Complaint ID, room, resolution description and completed date.
Daily pending reminder	Warden	Complaint ID, student, room, category, current status and complaint date.

5.8 User Management and Role-Based Access

The application defines four user roles: Student, Warden, Maintenance Staff and Admin. Role permissions should be configured at form, report, record and field level where supported. Student data visibility must be restricted using a reliable authenticated-user-to-student mapping. Hiding a report column alone is not sufficient to protect records. Test access using separate accounts for each role.

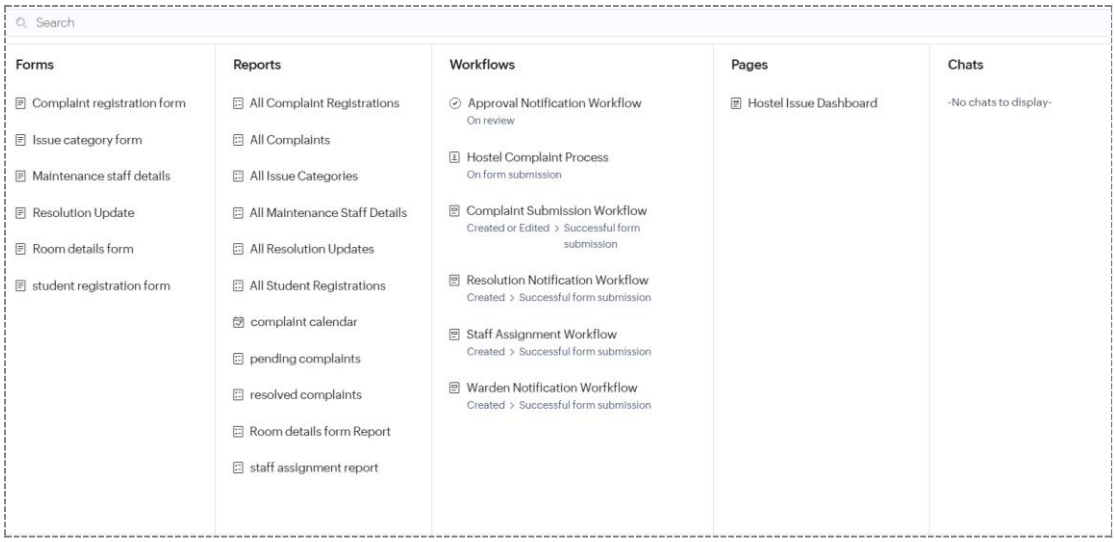


Figure 5.8: Role Hierarchy

5.9 Blueprint and Approval

The planned Blueprint, named “Hostel Complaint Process”, defines the states Submitted, Pending Approval, Approved, Rejected, Assigned, In Progress, Resolved and Closed. Transitions include Send for Approval, Approve, Reject, Assign Staff, Start Work, Mark Resolved and Close Complaint. Each transition should be restricted to the appropriate role and require relevant information.

The separate approval process, named “Warden Complaint Approval”, should route submitted complaints to the warden. The warden must make the decision; automatic approval must not be enabled. If the Creator edition does not provide Blueprint or approval functionality, record the limitation and use permitted manual status controls with restricted permissions.

5.10 Portal and Daily Schedule

A private student portal is planned so students can submit complaints and view only their own records. Portal user availability depends on the Zoho Creator plan and configuration. If portal access is unavailable, use authenticated application users and document the limitation. A daily schedule named “Pending Complaint Reminder” should search for complaints whose status is not Resolved, Rejected or Closed and send a summary to the warden. The schedule must avoid sending reminders for completed or rejected complaints.

CHAPTER 6

TESTING AND DEBUGGING

6.1 Testing Strategy

Testing should verify form validation, unique identifiers, lookup relationships, status changes, approval decisions, staff assignment, resolution updates, email notifications, reports, dashboard counts and role-based access. Each test should be performed in the live application using test records. Expected results are listed below; actual results and pass/fail status must be filled only after execution.

6.2 Test Cases

Table 6.1: Test Cases

Test ID	Test Procedure	Expected Result	Actual Result / Status
TC-01	Submit Student Registration with all required fields.	Student record saves and Student ID is generated.	To be tested
TC-02	Submit duplicate Register Number.	Duplicate record is rejected by unique validation.	To be tested
TC-03	Create a Room Details record.	Room ID is generated and required room details are stored.	To be tested
TC-04	Create an Issue Category record.	Category ID is generated and category/priority are stored.	To be tested
TC-05	Submit complaint with valid Student, Room and Category lookups.	Complaint saves with unique Complaint ID and Submitted status.	To be tested
TC-06	Submit complaint without required fields.	Mandatory validation prevents submission.	To be tested
TC-07	Submit complaint with an image.	Image is attached to the correct complaint record.	To be tested
TC-08	Warden approves a pending complaint.	Status changes to Approved and student notification is initiated.	To be tested
TC-09	Warden rejects a pending complaint with reason.	Status changes to Rejected and reason is retained; student is notified.	To be tested
TC-10	Assign an available maintenance staff member.	Status changes to Assigned and staff receives notification.	To be tested
TC-11	Maintenance staff starts work.	Status changes to In Progress for the assigned complaint.	To be tested
TC-12	Submit valid Resolution Update.	Resolution is linked to complaint and status changes to Resolved.	To be tested

Test ID	Test Procedure	Expected Result	Actual Result / Status
TC-13	Open All Complaints and Pending Complaints reports.	Records and filters match the underlying complaint data.	To be tested
TC-14	Open dashboard after adding test complaints.	KPI and chart values reflect current application data.	To be tested
TC-15	Student attempts to view another student's complaint.	Access is denied or record is not displayed.	To be tested
TC-16	Run daily pending reminder schedule.	Warden receives only unresolved/in-scope complaint reminders.	To be tested

6.3 Debugging Deluge Scripts

During debugging, verify the exact form and field link names, lookup return values, null checks, date comparisons, recipient addresses and workflow execution order. Use Creator's workflow history and execution details where available. A script should be tested with valid data, missing values, invalid relationships and duplicate submissions. Avoid inserting a new record when the current form record can be updated directly.

6.4 Defect Log

Table 6.2: Defect Log

Defect ID	Possible Issue	Risk	Recommended Correction
BUG-01	Duplicate complaint created by both form submission and workflow insert.	Duplicate records and incorrect dashboard counts.	Use the current record; do not insert a second complaint.
BUG-02	Lookup field returns an unexpected value or wrong display field.	Incorrect student, room or category association.	Verify lookup source form, display field and record ID.
BUG-03	Complaint status is updated before resolution validation completes.	Complaint may be marked resolved without valid evidence.	Validate and save resolution first, then update linked complaint.
BUG-04	Student report exposes records belonging to other students.	Privacy breach.	Apply record-level criteria tied to authenticated user identity and test with separate accounts.
BUG-05	Daily reminder includes resolved/rejected/closed complaints.	Unnecessary notifications.	Use explicit status exclusion criteria and test all status values.

Defect ID	Possible Issue	Risk	Recommended Correction
BUG-06	Email action uses an invalid or unconfigured recipient.	Notification failure.	Verify sender configuration, recipient address and email delivery logs.

6.5 User Acceptance Testing

User acceptance testing should be carried out with representative student, warden and maintenance staff accounts. The testers should complete the main complaint lifecycle and confirm that the screens, status labels, notifications and reports are understandable. Record tester feedback, date, test account role and final acceptance decision before submission.

CHAPTER 7

DEPLOYMENT, SECURITY AND GOVERNANCE

7.1 Environments and Deployment Strategy

The application should first be configured and tested in the available development or trial environment. Before sharing a live link, verify the application name, permissions, email settings, portal access, schedule configuration and data visibility. Production deployment status must be reported accurately; a trial or development application should not be described as a production deployment.

Table 7.1: Deployment Checklist

Checklist Item	Evidence / Status
Application name	Hostel Room Issue Tracker
Zoho Creator edition	To be confirmed from account
Application link name	To be copied from Creator
Live application URL	To be added after publication / verification
Development testing	To be completed and recorded
Production deployment date	Not confirmed
Backup / export	Follow the available Creator export and backup options

7.2 Performance Optimisation

Reports should retrieve only the fields needed for each view. Search and filter criteria should use appropriate fields, and workflow scripts should avoid repeated full-table retrievals. Dashboard charts should use relevant complaint fields and avoid unnecessary duplicate components. Performance should be checked with a realistic number of rooms, students and complaints.

7.3 Security Features

- Use authenticated users and assign only the minimum required role permissions.
- Restrict students to their own complaint records using record-level criteria or supported portal permissions.
- Restrict approval and assignment transitions to wardens or authorised administrators.

- Restrict work-progress and resolution actions to assigned maintenance staff and administrators.
- Protect student email addresses, phone numbers and complaint attachments from unauthorised access.
- Use verified email recipients and supported platform connections.
- Review application sharing, portal settings and permission profiles before publication.
- Do not store passwords or sensitive credentials in form fields or Deluge scripts.

7.4 Governance and Administration

The administrator should maintain student and room master data, review staff availability, check issue category values, monitor pending complaints and verify that notifications and schedules are active. Role assignments should be reviewed periodically. Workflow changes should be tested before being applied to the live application. Complaint and resolution records should be retained according to institutional policy.

CHAPTER 8

RESULTS AND DISCUSSION

8.1 Application Output

The expected application output is a centralised complaint-management interface in which students can submit room issues and track progress, wardens can review and assign complaints, and maintenance staff can update work and resolution details. The reports and dashboard are intended to provide a consolidated view of complaint volume, status, categories and staff assignments.

Actual screenshots, record counts, email delivery evidence and test results were not supplied with the reference requirements. Insert screenshots from the completed Zoho Creator application in this section and replace the verification notes with observed results. Do not report sample counts or successful tests unless they have been measured.

8.2 Achievement of Objectives

Table 8.1: Achievement of Objectives

Objective	Planned Evidence	Verification
Centralised data model	Six related forms and lookup relationships.	Verify forms and lookup behaviour.
Complaint tracking	Unique Complaint ID and eight status values.	Submit and track a test complaint.
Approval	Warden approval/rejection process and reason capture.	Test both decisions.
Staff assignment	Staff lookup, assignment date and notification.	Assign a test complaint.
Resolution tracking	Resolution Update with dates, description, image and feedback.	Submit a test resolution.
Reports and dashboard	Five reports and one dashboard with KPIs and charts.	Compare output with stored records.
Security	Student, Warden, Maintenance Staff and Admin roles.	Test each role with separate accounts.
Reminder and email	Event notifications and daily pending schedule.	Verify actual email delivery.

8.3 Comparison with Existing System

Table 8.2: Existing System versus Proposed System

Area	Manual Process	Proposed Application
Complaint records	Messages, notebooks or spreadsheets.	Centralised complaint form and unique ID.
Issue classification	Informal descriptions.	Configured issue categories and priority.
Approval	Verbal or message-based decision.	Structured warden review and decision.
Assignment	Manual forwarding to staff.	Linked staff assignment record.
Progress tracking	Follow-up messages.	Status lifecycle and resolution updates.
Evidence	Images stored separately in chats.	Complaint and resolution image fields.
Reporting	Manual filtering and counting.	List, filtered and calendar reports with dashboard.
Access	Shared records may expose information.	Role-based and record-level access controls.

8.4 Limitations of the Low-Code Approach

The application depends on Zoho Creator edition capabilities and configured account permissions. Blueprint, approval, portal users, schedules and some email actions may have plan-specific limits. Complex record-level security requires careful configuration and testing. The proposed staff performance summary depends on consistent assignment and completion data. The current design does not include predictive maintenance, IoT sensor integration, SMS/WhatsApp messaging or external hostel ERP integration.

8.5 Reflection on the No-Code → Low-Code → Pro-Code Spectrum

The project is primarily a low-code application. Forms, lookup relationships, reports, dashboard components, permissions, approval routing and Blueprint transitions are configured visually. Deluge can extend the application with validation, conditional updates and email notifications. The project demonstrates how visual configuration and targeted scripting can be combined to implement a practical administrative workflow.

CHAPTER 9

CONCLUSION AND FUTURE ENHANCEMENTS

9.1 Conclusion

Hostel Room Issue Tracker is designed to improve the way hostel room maintenance complaints are recorded, reviewed, assigned and resolved. The proposed Zoho Creator application centralises student, room, issue category, complaint, maintenance staff and resolution information through six related forms. A defined complaint lifecycle supports submission, warden review, approval or rejection, staff assignment, work progress, resolution and closure.

Five reports and a dashboard are planned to provide visibility into complaint records, pending work, resolved issues, staff assignments and complaint dates. Workflow notifications and a daily reminder are intended to reduce manual follow-up. Role-based permissions are designed to protect student information and separate the responsibilities of students, wardens, maintenance staff and administrators.

The project demonstrates the use of a low-code platform to model a real administrative process. The final level of achievement must be established through live application verification, screenshots and completed test cases. Features unavailable in the selected Creator edition should be documented transparently.

9.2 Future Enhancements

- Add SMS, WhatsApp or mobile push notifications where supported and approved.
- Introduce escalation rules for complaints that remain pending beyond a defined time.
- Add preventive maintenance schedules for hostel rooms and shared facilities.
- Provide QR-code scanning for room identification and faster complaint submission.
- Integrate with a hostel ERP or institutional student information system.
- Add analytics to identify recurring faults by room, block or issue category.
- Support multilingual labels for students and maintenance staff.
- Add maintenance inventory tracking for spare parts and repair materials.

- Provide service-level targets and overdue complaint indicators.
- Evaluate Zia or analytics capabilities for trend summaries after reliable data is collected.

9.3 Learning Outcomes and Career Relevance

The project provides practical experience in low-code application development, relational data modelling, lookup configuration, form validation, reports, dashboards, workflow automation, Deluge scripting, approval routing, Blueprint and role-based access control. These skills are relevant to business application development, workflow automation, application support, systems analysis and citizen-development roles. The project also highlights the importance of testing permissions, data integrity and automation before deployment.

REFERENCES

[1] Zoho Corporation, “Zoho Creator Help Documentation,” [Online]. Available: https://www.zoho.com/creator/help/. Accessed: 10-Nov-2026.

[2] Zoho Corporation, “Deluge Script Reference,” [Online]. Available: https://www.zoho.com/deluge/. Accessed: 10-Nov-2026.

[3] Zoho Corporation, “Blueprint in Zoho Creator,” [Online]. Available: https://www.zoho.com/creator/help/. Accessed: 10-Nov-2026.

[4] Zoho Corporation, “Approval Process in Zoho Creator,” [Online]. Available: https://www.zoho.com/creator/help/. Accessed: 10-Nov-2026.

[5] Zoho Corporation, “Portal Users and Permissions in Zoho Creator,” [Online]. Available: https://www.zoho.com/creator/help/. Accessed: 10-Nov-2026.

[6] Zoho Corporation, “Schedules and Email Notifications in Zoho Creator,” [Online]. Available: https://www.zoho.com/creator/help/. Accessed: 10-Nov-2026.

APPENDIX A

ZOHO CREATOR FEATURE COVERAGE CHECKLIST

Table A.1: Feature Coverage Checklist

Feature	Planned Coverage	Evidence to Attach
Forms and lookups	6 forms; complaint relationships.	Screenshots of all six forms and lookup settings.
Auto Number and validation	Unique IDs and required fields.	Form configuration and sample records.
Reports	5 reports including list and calendar.	Screenshots of each report.
Dashboard	4 KPIs, 2 charts, recent complaints and staff summary.	Dashboard screenshot with sample data.
Deluge / workflows	5 workflow automations as applicable.	Workflow configuration and execution logs.
Blueprint	8 states and defined transitions.	Blueprint screenshot and transition tests.
Approval	Warden approval and rejection.	Approval configuration and decision test.
Schedule	Daily pending complaint reminder.	Schedule configuration and email evidence.
Email integration	Warden, student and staff notifications.	Test emails / execution history.
RBAC / portal	Student, Warden, Maintenance Staff and Admin roles.	Permission matrix and role test evidence.

Mark each item as Implemented, Partially Implemented, Not Available or Not Tested only after verifying the actual Creator application.

APPENDIX B

COURSE OUTCOME MAPPING

Table B.1: Mapping of Course Outcomes

CO	Course Outcome Area	Evidence in Report
CO1	Explain principles, architecture and business value of low-code application development.	Chapters 1 and 2
CO2	Develop business applications using forms, reports, workflows and Deluge scripting.	Chapters 4, 5 and 6
CO3	Configure integrations, notifications and role-based security.	Chapters 5 and 7
CO4	Deploy, secure and manage applications using governance practices.	Chapter 7
CO5	Evaluate low-code development approaches and application limitations.	Chapter 8 and Chapter 9

Confirm the exact course outcome statements with the course faculty and replace this mapping if the official syllabus uses different wording.

APPENDIX C

SAMPLE DELUGE LOGIC AND CONFIGURATION NOTES

The following are pseudocode-level examples. They are provided to explain the intended business logic and must be converted into valid Deluge using the exact field link names and workflow context from the live application.

C.1 Complaint Submission

```
// Configure Complaint ID as an Auto Number field.
// Configure Status default value as "Submitted".
// Do not insert a second Complaint Registration record
// from the on-success workflow.
```

C.2 Status and Resolution Validation – Logic Outline

```
// Pseudocode only – adapt field link names before use.
if(input.Complaint == null)
{
    alert "Please select a complaint.";
    cancel submit;
}

if(input.Assigned_Staff == null)
{
    alert "Please select the assigned staff member.";
    cancel submit;
}

// After successful resolution validation:
// retrieve the linked complaint record;
// update its Status to "Resolved";
// send the student notification.
// Ensure the workflow does not create duplicate records.
```

Important: A lookup field may return a record ID or a record object depending on context. Verify the actual Deluge syntax using the Creator editor and test with sample records. Do not paste pseudocode directly into a live workflow.

APPENDIX D

USER MANUAL

D.1 Accessing the Application

- Open the verified Hostel Room Issue Tracker application URL.
- Sign in using the account assigned to the user's role.
- Open the navigation menu and select the form, report or dashboard available to that role.
- Do not share login credentials with other users.

D.2 Instructions for Students

1. Open the Complaint Registration form.
2. Select your student record and the correct hostel room.
3. Choose the issue category and enter a clear complaint title and description.
4. Select the priority and attach an image if it helps explain the issue.
5. Submit the complaint and note the generated Complaint ID.
6. Open the permitted complaint report to track status and view available resolution details.

D.3 Instructions for Wardens

1. Open the Pending Complaints report.
2. Review the student, room, category, priority, description and uploaded image.
3. Move the complaint through the approval process.
4. Approve or reject the complaint; provide a rejection reason where required.
5. For approved complaints, select an appropriate maintenance staff member and record the assignment date.
6. Monitor progress through the All Complaints and Staff Assignment reports.

D.4 Instructions for Maintenance Staff

1. Open the assigned complaints view available to the staff role.
2. Review the room, category, priority and complaint description.

3. Start work and update the complaint progress using the permitted transition.
4. Complete the Resolution Update form with work description and relevant dates.
5. Upload a resolution image if required and submit the update.
6. Ensure that the complaint status is updated after valid resolution submission.

D.5 Instructions for Admin

1. Maintain student, room, category and maintenance staff master records.
2. Check role assignments and record-level permissions.
3. Verify email sender settings, recipient addresses and notification actions.
4. Review the daily pending reminder schedule.
5. Test workflow changes and retain application backup/export evidence where available.

APPENDIX E

EVALUATION RUBRIC – FOR EXAMINER USE

Student	Register No.	Project Title
M. Siva Swathi	240701516	Hostel Room Issue Tracker

Table E.1: Evaluation Rubric for the Individual Project

S.No.	Criterion	Indicators	Max. Marks	Marks Awarded
1	Problem Definition & Requirement Analysis	Problem clarity, objectives, functional and non-functional requirements, use cases.	10	
2	System Design	Architecture, ER/data model, form specifications, workflow and permission design.	10	
3	Forms, Reports & Pages	Field types, lookup relationships, reports, dashboard and UI.	15	
4	Deluge Scripting & Workflows	Validation, notifications, schedules, approval and Blueprint.	15	
5	Integration, RBAC & Security	Email integration, roles, permissions and data privacy.	10	
6	Testing & Debugging	Test coverage, defect handling and evidence.	10	
7	Documentation & Presentation	Report quality, screenshots, user manual and explanation.	10	
8	Demonstration & Viva	Working demonstration and response to questions.	20	
	TOTAL		100	